Transformation of *E. coli* (heat-shock and electroporation)

What you're going to get / What you expect

E. coli colonies carrying the construct on selective plates, first in a cloning strain (DH5a) and then in a (DE3) expression host, together with recovery of clean plasmid by miniprep.

Materials list (per transformation)

- Plasmid DNA or ligation dialysate.
- Chemo-competent or electro-competent *E. coli* (DH5a for cloning; a (DE3) strain for expression).
- LB-SOC medium (Luria-Bertani plus Super Optimal broth with Catabolite repression).
- Water bath at 42 C; ice; 0.2 cm electroporation cuvette and electroporator (for the electro route).
- LB-agar plates with the relevant antibiotic (kanamycin for the pET-28a series, ampicillin for the pET-22b series).
- A miniprep kit (alkaline lysis, e.g. NZY Miniprep).

Steps

Chemo-competent route

1. Aliquot 1 uL of DNA onto 50 uL of cells and incubate 30 minutes on ice.
2. Apply heat shock in a water bath at 42 ºC for 90 seconds, return to ice for 5 minutes, and add 0.9 mL of LB-SOC.

Electro-competent route

1. Aliquot 1 uL of DNA onto 30 uL of cells, incubate 1 minute on ice, transfer to a pre-cooled 0.2 cm cuvette and apply a pulse at 25 uF, 2.5 kV and 200 ohm (optimal time constant 5 to 5.5 ms); immediately add 1 mL of LB-SOC.

For both routes

1. Recover the cells for 1 hour at 37 ºC so they express the antibiotic resistance.
2. Plate a sufficient volume on LB-agar with the relevant antibiotic and incubate overnight at 37 ºC.
3. Clone in DH5a first; verify positive colonies (colony PCR, see the PCR protocol), recover their plasmid by alkaline-lysis miniprep, and re-transform the verified plasmid into the (DE3) expression strain.

Troubleshooting

- Plating straight onto antibiotic without the 1 hour recovery kills transformed-but-not-yet-resistant cells: always recover in LB-SOC first.
- Clone in DH5a first (high plasmid yield, no T7 polymerase) and only move the verified plasmid into the (DE3) host afterwards.